INCH-POUND

MIL-T-82825(OS)
29 December 1989
SUPERSEDING
WS 20883
8 November 1983
(See 6.7)

MILITARY SPECIFICATION

TRIPHENYLBISMUTH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one grade of triphenylbismuth, referred to herein as triphenylbismuth (TPB) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation
(see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1376

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-5094.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Chemical and physical properties. Chemical and physical properties of triphenylbismuth shall conform to the requirements specified in table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Purity, percent	97.0	---
Melting point, degrees Celsius (°C)	76.0	---
Acetone insoluble matter, percent	---	0.10
Magnesium, percent	---	0.003
Total halides as chloride, percent	---	0.05

3.3 Safety.

3.3.1 Toxic products and formulations. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 Material Safety Data Sheet (MSDS). The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.4 Shelf life. The triphenylbismuth shall have a shelf life of 12 months after date of acceptance testing when stored at ambient temperatures and in the absence of ultraviolet light.

3.4.1 Shelf life extension. The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the triphenylbismuth, upon retest conforms to the purity, melting point, and acetone insoluble requirements.

3.5 Workmanship. The triphenylbismuth shall be uniform in appearance and free of impurities and foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

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4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspections.

Examination and tests	Requirement	Test method
Visual examination	3.5 and Section 5	4.5.1
Purity test	3.2	4.5.2
Melting point test	3.2	4.5.3
Acetone insoluble matter test	3.2	4.5.4
Magnesium test	3.2	4.5.5
Total halides test	3.2	4.5.6

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of all of the triphenylbismuth subject to inspection at one time from one or more batches or a homogeneous blend of all batches of triphenylbismuth manufactured using the same set of materials and processing operations and conditions and submitted for acceptance at one time. If all the batches in the lot were mixed to make a homogeneous blend, the lot shall be considered to consist of one batch for purpose of sampling and testing.

4.4.1.2 Selection of sample containers. Sample containers of the triphenylbismuth shall be selected at random from each lot for the inspections specified in 4.5.1 through 4.5.6. Sample material shall be selected as specified in 4.4.1.3. Sampling for tests shall consist of one 50-gram (g) minimum sample selected at random from each batch (see 6.5.1). The batch samples shall not be blended. All the tests shall be performed on each sample, and test results shall be recorded for each batch. Sampling to determine conformance to section 5 shall consist of all the containers in the lot.

4.4.1.3 Sample material. Sufficient material shall be selected at random from each sample container to perform all the inspections specified in 4.5.1 through 4.5.6.

4.4.1.4 Noncompliance of samples. Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

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4.5 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 Visual examination. Each sample of material shall be visually examined to determine conformance to 3.5.

4.5.2 Purity test. The purity of triphenylbismuth shall be determined as follows:

4.5.2.1 Apparatus.

a. Liquid chromatograph equipped as follows:

(1) Detector. Ultraviolet (UV) variable wave-length with LC flow-thru assembly.

(2) Column C18. Bonded reversed phase, 5-10 micrometers.

(3) Pump. Capable of operating up to 4000 psi pressure.

b. Laboratory glassware.

4.5.2.2 Reagents and standards.

a. Triphenylbismuth. Recrystallized from heptane to maximum absorbance.

b. Triphenylmethane. Reagent grade.

c. Water. High pressure liquid chromatography (HPLC) grade.

d. Methanol. HPLC grade.

e. Acetone. UV grade.

f. Acetonitrile. HPLC grade.

4.5.2.3 Procedure.

a. Prepare a stock solution of internal standard triphenylmethane (TPM) by weighing 0.5 $\pm$ 0.1-g of TPM in a 100-milliliter (ml) volumetric flask and diluting to the mark with 50 ml acetone and 50 ml acetonitrile.

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b. Weigh 0.5 ± 0.1 g of TPB sample to be analyzed to the nearest 0.1 milligram (mg) in each of three 100-ml volumetric flasks and dilute to the mark with 50 ml acetone and 50 ml acetonitrile.

c. Weigh 0.5 ± 0.1 g of TPB standard to the nearest 0.1 mg in each of two 100 ml volumetric flasks and dilute to the mark with 50 ml acetone and 50 ml acetonitrile.

d. Accurately transfer a 1.0-ml aliquot from each of the 100-ml flasks into five separate 25-ml volumetric flasks.

e. Accurately pipette 1.0 ml of the TPM internal standard stock solution in each of the five 25-ml flasks. Dilute to the mark with 50:50 acetone and acetonitrile.

f. Analyze the three samples and two standard solutions using a liquid chromatograph. The following instrument parameters have been found satisfactory in this analysis.

(1) Detector. UV variable wavelength, such as Varian UV 50, set at 226 nanometers.

(2) Eluent. 80 percent methanol - 20 percent water.

(3) Flow rate. Approximately 2.0 ml/minute.

(4) Injection volume. Use the same injection volume for sample and standard solutions.

Adjustment of these parameters may be necessary to provide proper peak separation and quantification.

g. Determine the areas under both TPM and TPB peaks for each sample and standard.

h. The internal standard corrected response factors for the standards should agree within 2 parts per hundred. If not, repeat the test.

Calculate the average response factor:

Average response factor (R) = $1/2\ [W_1(I_1/A_1) + W_2(I_2/A_2)]$

Where A = area of standard TPB peak.

W = standard concentration, mg/ml.

I = area of standard TPM peak.

1 = standard 1 I_1/A_1

2 = standard 2 I_2/A_2

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i. Calculate the percent TPB in each sample as follows:

$$\text{Percent TPB} = \frac{R \times B}{C \times Is} \times 100$$

Where: R = response factor defined above.

B = area of sample TPB peak.

Is = area of sample TPM peak.

C = sample concentration, mg/ml.

j. Record TPB as the average of three determinations, to the nearest 0.1 percent.

4.5.3 Melting point test. The melting point of the triphenylbismuth shall be determined as follows:

4.5.3.1 Apparatus.

a. Capillary melting point apparatus - Electro-thermal, Mettler, Vanderkamp, or similar apparatus.

b. Rheostat control.

c. Capillary tubes - approximately 3 inches long.

d. Thermometer - covering desired range with $0.2^{0}C$ or smaller subdivision.

4.5.3.2 Procedure.

a. Reduce the sample to a very fine powder, and desiccate for 4 hours.

b. Completely seal one end of a capillary tube. Charge the tube by taking a watch glass covered with test material and pressing the open end of the tube against the watch glass.

c. Closely pack the material in the bottom of the capillary to a depth of 3.0 $\pm$ 0.6-mm by repeatedly dropping the capillary inside a 12-inch length of 1/4-inch OD glass tubing which is held vertically with the lower end about 1 inch above the table top.

d. Set up the Vanderkamp melting point apparatus near a standard electrical outlet with two receptacles. Plug in the light cord directly, and connect the heater through the rheostat control.

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e. Inset a thermometer into the hole of the top center of the Vanderkamp apparatus, and lower the thermometer all the way in. Lower one capillary tube into one of the small holes near the thermometer hole.

f. Heat until 25°C below the expected melting temperature, then carefully regulate the rate of temperature rise by means of the rheostat to about 3°C per minute until the material begins to melt. (This is the temperature at which the sample is observed to collapse definitely against the sides of the tube at any point.)

g. From this time, carefully regulate the temperature rise to 1/2 degree/minute until the material becomes liquid throughout.

h. Record the temperature at which the material completely melts as the melting point.

4.5.4 Acetone insoluble matter test. Acetone insoluble matter shall be determined as follows:

a. Weigh, to the nearest mg, approximately 10 g of sample into a 600-ml beaker, and dissolve in 400 ml of technical grade acetone.

b. Filter the solution through a tared filtering crucible, rinsing out the beaker with acetone and passing the rinsings through the crucible.

c. Wash the residue in the crucible with acetone until the washings are colorless.

d. Dry the crucible at 100 ± 5°C for 1 hour, cool in a desiccator and weigh.

e. Calculate the acetone insoluble matter as follows:

$$\text{Acetone insoluble matter, percent} = \frac{(B - A)}{W} \times 100$$

Where: A = tare weight of filtering crucible, g.

B = weight of crucible and residue, g.

W = weight of sample, g.

4.5.5 Magnesium content test. Magnesium content shall be determined as follows:

4.5.5.1 Apparatus.

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a. Atomic absorption spectrometer.

b. Hollow cathode lamp - magnesium.

c. Burner - 4-inch single slot for air-acetylene.

4.5.5.2 Reagents.

a. Nitric acid - chemically pure (C.P.) grade.

b. Perchloric acid - C.P. grade.

c. Hydrochloric acid - C.P. grade.

d. Magnesium standard solutions - Weigh 1.6920 g of magnesium oxide, ACS reagent grade, into a 1-liter volumetric flask, dissolve in 5 ml of hydrochloric acid, and make to volume with distilled water. One ml of this solution contains 1000 micrograms of magnesium. Transfer 10 ml of the solution into a 100-ml volumetric flask, and make to volume. One ml of this solution contains 100 micrograms of magnesium. Prepare working standards daily containing 3 and 6 micrograms/ml magnesium by placing appropriate aliquots of the 100micrograms/ml magnesium into 100-ml volumetric flasks and diluting to volumes with distilled water.

4.5.5.3 Standardization.

a. Determine absorbance value for each standard solution using the following atomic absorption conditions:

(1) Wavelength - 285.2 nm.

(2) Slit setting - Follow instrument manufacturer's instructions.

(3) Light source - Hollow cathode lamp, magnesium.

(4) Flame - air-acetylene.

b. Run a reagent blank.

c. Prepare a calibration curve by plotting the absorbance corrected for the reagent blank reading against the concentration of magnesium for each standard.

4.5.5.4 Procedure.

a. Weigh 1.0 g of triphenylbismuth in a 100-ml beaker.

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b. Add 10 ml of nitric acid and then add 5 ml of perchloric acid, and carefully digest the sample.

c. Transfer the clear sample to a 25-ml volumetric flask, and make to volume with distilled water.

d. Determine the absorbance value using the same conditions as for the standards.

e. Run a reagent blank the same as the samples, and subtract this intensity from that of each sample.

f. Determine the concentration of magnesium from the calibration curve.

g. Calculate the percent magnesium as follows:

$$\text{Percent magnesium content} = \frac{A \times D \times 10^{-6}}{W} \times 100$$

Where: A = concentration of magnesium in sample solution, micrograms/ml.

D = sample dilution volume, 25 ml.

W = sample weight, g.

10^6 = unit conversion factor (micrograms to grams).

4.5.6 Total halides as chloride test. Total halides as chloride shall be determined as follows:

4.5.6.1 Apparatus.

a. Laboratory glassware.

b. Magnetic stirrer.

4.5.6.2 Reagents.

a. Nitric acid - chemically pure (C.P.) grade.

b. Acetone - technical grade.

c. Chloride standard solution - Weigh 0.165 g of sodium chloride in a 1-liter volumetric flask, and dissolve in and make to volume with distilled water. One ml of this solution contains 0.1 mg of chloride.

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d. Silver nitrate solution - Dissolve 1.7 g of silver nitrate in 100 ml of distilled water.

4.5.6.3 Procedure.

a. Weigh 0.20 g of triphenylbismuth in a 50-ml beaker.

b. Add 25 ml of distilled water and 2 ml of acetone.

c. Mix on a magnetic stirrer for 15 minutes.

d. Filter the sample solution in a 50-ml beaker.

e. Add 1 ml of nitric acid and 1 ml of silver nitrate solution.

f. Prepare a blank using 1 ml of standard Cl solution, and compare with the sample.

g. After 5 minutes, if the turbidity of the sample is less than or the same as the blank, record the sample as containing not more than 0.05 percent total halides as chloride. If the turbidity of the sample is greater than the blank, record the sample as containing more than 0.05 percent total halides as chloride.

4.6 Inspection of packaging. The TPB containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The triphenylbismuth shall be packaged to afford adequate protection against loss and contamination during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified. Containers in the same shipment shall be of the same size. Unless otherwise specified in the procurement document, the triphenylbismuth shall be packaged in fiberboard drums. Net weight in each drum shall be not greater than 50 pounds. Each drum shall be lined with a close-fitting polyolefin bag liner. The bag liner shall be closed by tying. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

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5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension (see 3.4). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ADDITIONAL 12 MONTH PERIODS PROVIDED THE TRIPHENYLBISMUTH, UPON RETEST, CONFORMS TO THE PURITY, MELTING POINT, AND ACETONE INSOLUBLE REQUIREMENTS OF MIL-T-82825."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The triphenylbismuth specified herein is intended for use as an ingredient in Standard Missile Mark 104 Dual Thrust Rocket Motor solid propellant.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheet (see 3.3.2 and 6.4).

e. Packaging required, if different than 5.1.1.

f. Special marking, if other than as specified in 5.2.2.

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6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the triphenylbismuth sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Material Safety Data Sheet (MSDS). MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MDSDs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Definition.

6.5.1 Batch. A batch shall consist of all the triphenylbismuth manufactured at the same time in one reaction unit using the same set of materials and processing operations and conditions.

6.6 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mark 104
Solid propellant
Standard Missile

6.7 Supersedure information. MIL-T-82825 incorporates the following engineering change proposal (ECP), and notice of revision (NOR):

ECP	NOR
MTA066 (5/5/88)	MTA066.1 (3/24/88)

Preparing Activity:
NAVY-OS
(Project 1376-N366)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82825(OS)	2. DOCUMENT DATE (YYMMDD) 891229

3. DOCUMENT TITLE
Triphenylbismuth

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-744-4700 (2) DSN: 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office, Attn: DLSC-LM 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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